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Bilateral asymmetry of quantitative parenchymal kinetics at ultrafast DCE-MRI predict response to neoadjuvant chemotherapy in patients with HER2+ breast cancer

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Abstract

Purpose: To assess whether measurement of the bilateral asymmetry of semiquantitative and quantitative perfusion parameters from ultrafast dynamic contrast-enhanced MRI (DCE-MRI), allows early prediction of pathologic response after neoadjuvant chemotherapy (NAC) in patients with HER2+ breast cancer.

Materials and Methods: Twenty-eight female patients with HER2+ breast cancer treated with NAC who underwent pre-NAC ultrafast DCE-MRI (3 – 9 seconds/phase) were enrolled for this study. Four semiquantitative and two quantitative parenchymal parameters were calculated for each patient. Ipsilateral/contralateral (I/C) ratio (for four parameters) and the difference between (for two parameters) ipsi- and contra-lateral parenchymal kinetic parameters (kBPE) were compared for patients with pathologic complete response (pCR) and those having residual

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disease. *Lasso regression with leave-one-out cross validation was used to determine the optimal combination of parameters for a regression model and multivariable logistic regression was used to identify independent predictors for pCR. Chi-squared test, two-sided t-test and Kruskal-Wallis test were used.*

Results: The K^{trans} I/C ratio cutoff value of 1.11 had a sensitivity of 83.3% and specificity of 75% for pCR. The v_e I/C ratio cutoff value of 1.1 had a sensitivity of 75% and specificity of 81.3% for pCR. The area under the receiver operating characteristic curve of the three-kBPE parameter model, including initial area under the enhancement curve (AUC_{30}) I/C ratio, K^{trans} I/C ratio and v_e I/C ratio, was 0.89 with sensitivity of 91.7% at specificity of 81.3%.

Conclusion: Quantitative assessment of bilateral asymmetry kBPE from pre-NAC ultrafast DCE-MRI can predict pCR in patients with HER2+ breast cancer.

Keywords

ultrafast DCE-MRI; HER2+ breast cancer; response to neoadjuvant chemotherapy; BPE; bilateral asymmetry

1. Introduction

Neoadjuvant chemotherapy (NAC) is a well-established treatment for breast cancer, which downstages the tumor for a less-invasive surgery and allows informed adjustment of the treatment plan [1]. Achieving a pathologic complete response (pCR; i.e. the absence of residual invasive cancer or local lymph nodes) following NAC is strongly associated with improved long-term disease-free and overall survival [2]. Compared with most other subtypes of breast cancer, a higher percentage of human epidermal growth factor receptor 2 positive (HER2+) patients achieve a pCR to NAC with an averaged pCR rate of 36.4% (ranged from 17.5% to 74.2%) [3]. Unfortunately, most patients still have residual disease at the time of surgery, representing a higher-risk population compared with patients who achieve pCR. In the past decades, pathologic response has been used as a gold standard to guide further adjuvant treatment [4]. However, pCR can only be confirmed from surgical resection specimens [5], which may have sampling error [6]. Moreover, an accurate prediction of pathologic response pre-NAC or early during NAC has diagnostic and prognostic values, because it could spare non-responders long- and short-term toxicities from ineffective chemotherapies [7] and guide the development of optimal NAC for each patient [8]. For example, physicians could select patients for escalation with more potent (but more toxic) antibody drug conjugates [9], based on their initial prognosis.

Breast dynamic contrast-enhanced MRI (DCE-MRI) is a reliable tool for detecting breast cancer before NAC [10] and can be used to predict response of treatment [1, 11]. Ultrafast DCE-MRI with high temporal resolution is an emerging protocol that allows accurate measurements of perfusion and tumor-associated vasculature based on contrast media uptake kinetics during the initial 60 seconds after contrast agent administration [12].

Previous studies reported that perfusion parameters from ultrafast DCE-MRI, such as the initial slope of enhancement in lesions and bolus arrival time in arteries and lesions, are strongly associated with breast cancer malignancy [12–14]. There have been few studies of

the role of ultrafast DCE-MRI in predicting the response to NAC [6, 7, 15–17]. Recently, Ramtohul *et al.* [15] found that the pre-NAC tumor wash-in slope assessment with ultrafast DCE-MRI may predict pCR, which was contradict with other earlier studies [7, 16]. In a study of response of triple negative breast cancer (TNBC) to NAC [17], functional tumor volume calculated from the early and late phase of ultrafast DCE-MRI acquired after cycle 4 of NAC was shown to predict pCR, but there was no evidence of association between the functional tumor volume from pre-NAC MRIs and pCR.

In the past decade, studies [18, 19] suggested that the enhancement of normal fibroglandular breast tissue, known as background parenchymal enhancement (BPE), could be a predictor of the response to NAC. Although decreased BPE as early as after the first or second cycle of NAC was found to be strongly linked to pCR [20], a consistent association between pre-NAC BPE and pCR has not been convincingly demonstrated [21–24]. Nevertheless, the bilateral difference of parenchymal kinetic parameters from ultrafast DCE-MRI, including maximum slope of enhancement (*MaxSlope*), bolus arrival time (*BAT*) and initial area under signal enhancement curve over 30 seconds (*AUC30*) were significantly associated with pCR [7].

In this study, we calculated both semi-quantitative and quantitative parameters from the pre-treatment ultrafast DCE-MRI to assess whether the pre-NAC bilateral asymmetry of parenchymal kinetic parameters (kBPEs) were associated with pCR after NAC in patients with HER2+ breast cancer. The kBPEs of patients with HER2+ breast cancer were also compared with those of patients with negative findings.

2. Material and Methods

2.1. Study sample

This retrospective study received institutional review board approval with a waiver of the requirement for written informed consent.

We retrospectively searched the institutional electronic medical records between December 2016 to November 2020 to identify patients who met the following criteria: (1) diagnosed with unilateral invasive ductal carcinoma (IDC) which was classified as HER2+ by ASCO/CAP guidelines (<https://www.silverchair.com>) and treated with NAC, (2) bilateral MRI prior to NAC, and (3) a hybrid clinical protocol that included ultrafast DCE-MRI for 60 – 80 s after contrast agent administration. As result, 33 female patients with HER2+ breast cancer were treated with NAC. Five patients were excluded because of no ultrafast MRI (n = 4) or bilateral tumors (n = 1). In addition to the patients with HER2+ breast cancer, we included 11 screening patients who met the inclusion criteria (3) as control cases. None of these screening patients developed malignant tumor within 1 year of their MRI.

Pre-NAC tumors were evaluated by histologic grading. Treatment outcome was graded by residual cancer burden (RCB) index, estimated from the surgical pathology of the residual primary breast tumor and regional lymph nodes at the completion of NAC. The RCB index is classified using four categories with an increasing amount of residual disease: RCB 0 (pCR), RCB-I, RCB-II, and RCB-III. All clinicopathologic data (age, histologic grade,

adjuvant therapy, pathologic responses to NAC, RCB index) were collected through review of medical records.

2.2. MRI Acquisitions

All patients underwent MRI examinations in the prone position on a Philips Achieva 3.0 T TX scanner (Philips Healthcare, Best, the Netherlands) with a 16-channel bilateral breast coil (Mammo Trak, Philips Healthcare). The acquisition protocol included T2-weighted, T1-weighted, diffusion-weighted, conventional and ultrafast DCE-MRI sequences in the axial orientation. The ultrafast DCE-MRIs were acquired using the following parameters: acquisition pixel size = $1.5 \times 1.5 \text{ mm}^2$; slice thickness = 3 – 4 mm; number of slices = 80 – 133; TR (repetition time) = 2.8 – 3.2 ms, TE (echo time) = 1.4 – 1.6 ms, field-of-view (FOV) = 320 – 360 mm; flip angle = 10° ; SENSE acceleration factor = 8. Temporal resolution ranged from 3 to 9 seconds, depending on the FOV and the number of slices. The ultrafast DCE-MRI protocol consisted of 5 pre-contrast phases and 8 – 19 post-contrast phases (starting 10 seconds after contrast agent injection and ending 60–80 seconds after injection). A dose of 0.1 mmol/kg of Gadavist (Bayer Healthcare Pharmaceutical, Berlin, Germany) or MultiHance (Bracco, NJ) was injected at 2 mL/s, followed by a saline flush of 20 mL at 2 mL/s.

2.3. Image analysis

Data analysis was performed with an in-house developed MATLAB platform (MathWorks, Natick, MA) and 3D Slicer (<http://www.slicer.org>) [25]. Ultrafast images were motion corrected by a non-rigid registration method [26]. A semi-automatic volumetric segmentation was performed with 3D Slicer using the first pre-contrast ultrafast time point. The segmentation process was detailed in previous study [7]. After segmentation, we obtained ipsilateral and contralateral normal parenchymal regions-of-interests (ROIs) which included all normal parenchymal tissue for each patient. Tumors were manually identified and segmented by a radiologist (12 years of experience in breast MRI) based on subtracted images (post-pre contrast image) and excluded from the parenchymal ROIs to yield ROIs that included only normal - uninvolved - parenchyma.

For each voxel within the normal parenchymal ROIs, the time course of percent signal enhancement, $PSE(t) = \frac{S_{post}(t) - S_{pre}}{S_{pre}} \times 100\%$, was computed from ultrafast DCE-MRIs, where $S_{post}(t)$ denotes the signal intensity at time ' t ', and S_{pre} is the average of the signal intensities from the five pre-contrast images. A digital filter was applied to select top 10% significantly enhancing voxels [14]. Top 10% ROI was defined as the set of parenchymal voxels in the ipsilateral or contralateral breast that had signal enhancements within the highest 10% of all normal parenchymal voxels according to $PSE(t)$ at the last time point of the ultrafast series. Enhancement kinetics for the ipsilateral and contralateral breasts were calculated separately.

2.4. kBPE parameters

We fit $PSE(t)$ to a truncated (wash-in only) empirical mathematical model:

$$PSE(t) = (t \geq t_0) \cdot A \cdot \frac{(\alpha(t-t_0))^2}{1 + (\alpha(t-t_0))^2}, \quad (1)$$

where t_0 is the bolus arrival time (*BAT*, s), A is the upper limit of percent enhancement, and α is the uptake rate (s^{-1}). The constraint, ($t \geq t_0$), is to make sure that there is no signal enhancement before contrast bolus arrival. An ROI within the aorta was selected to calculate the *BAT* in the aorta using Eq. (1). Parenchymal *BAT* was then calculated relative to the *BAT* in the aorta. Given the parameters estimated from Eq. (1), three secondary parameters were derived:

- i. Maximum relative enhancement slope (*MaxSlope*, in units of s^{-1}) is the first derivative of Eq. (1) at the time when its second derivative equals zero, i.e.,

$$MaxSlope = \frac{3\sqrt{3}}{8} A \cdot \alpha \quad (2)$$

- ii. Initial area under the curve, calculated by integrating the signal enhancement curve given by Eq. (1) over the initial $t = 30$ seconds (*AUC30*) post *BAT* in the voxel [12], i.e.,

$$AUC30 = A \cdot (t - \tan^{-1}(\alpha \cdot t)/\alpha) \quad (3)$$

- iii. Time to maximum slope (*TTMS*) that calculated from (i), i.e.,

$$TTMS = \frac{1}{\sqrt{3}\alpha} + t_0 \quad (4)$$

To calculate pharmacokinetic parameters of normal parenchymal tissue, signal intensity of voxel in parenchymal ROI and aorta ROI over the time of whole ultrafast series was converted into concentration of contrast media over time ($C(t)$) using the non-linear model of gradient echo signal equation [27] with literature baseline T1 values of lesion and blood of 1.445 s [28] and 1.544 s [29], respectively. The relaxivity of MRI contrast agent of $4.5 \text{ mMol}^{-1}\text{s}^{-1}$ for Gadavist and $5.4 \text{ mMol}^{-1}\text{s}^{-1}$ for Multihance [30] were used in the calculations.

Based on the standard Tofts model for DCE-MRI [31], changes in $C(t)$ for tissue following the contrast bolus injection are given by:

$$C(t) = K^{trans} \int_0^t C_p(\tau) e^{-\frac{K^{trans}}{v_e} \cdot (t-\tau)} d\tau$$

(5)

where K^{trans} refers to the transfer constant and v_e is the extravascular extracellular space (EES) fractional volume, $C_p(t) = \frac{C_b(t)}{1 - Hct}$ is the arterial input function (AIF), $C_b(t)$ is contrast agent concentration in blood, and Hct is the hematocrit, $Hct = 0.42$ was used in this study [32].

To quantify the bilateral asymmetry in normal parenchymal enhancement, the ipsilateral to contralateral ratio (I/C) for the parameters *MaxSlope*, *AUC30*, K^{trans} and v_e were calculated by using averaged across the ipsilateral normal parenchymal over the averaged value in the contralateral parenchymal. In addition, the difference between Ipsilateral and contralateral (C-I) for the parameters *TTMS* and *BAT* were calculated by subtracting the averaged value in contralateral parenchyma from the averaged value in ipsilateral normal parenchyma. An $I/C > 1$ or C-I positive means that a larger kBPE parameter was measured in the ipsilateral normal parenchyma compared to the contralateral parenchyma. Since there is no affected breast in the control cases, the ‘ipsilateral’ breast was randomly selected from the left or right breast.

2.5. Statistical Analysis

Correlation of baseline NAC patient data with pathologic response was performed with a chi-squared test for categorical variables and a two-sided t-test for continuous variables. The Kruskal-Wallis test was used for multiple comparison of kBPE parameters between control group, pCR and all residual diseases (non-pCR). We calculated odds ratios (ORs) with 95% confidence intervals (CIs) and p values for each parameter. The area under the receiver operating characteristic (ROC) curves were calculated, and optimal cutoff values to identify pCR were derived using the Youden index [33]. The optimal combination of kBPE parameters was selected using Lasso regression [34] by entering all bilateral kBPE asymmetry parameters. Multivariable logistic regression was performed using the selected parameters and validated by leave-one-out cross validation (LOOCV). A value of $p < 0.05$ was considered statistically significant.

3. Results

Twenty-eight patients ($n = 28$) with biopsy-proven IDC grade II ($n = 4$) or III ($n = 24$) were analyzed in this study. The baseline characteristics are summarized in Table 1. After completion of NAC, 12 patients (43%) had pCR and 16 (57%) had residual disease. No significant difference was found between the pCR and non-pCR groups in terms of age, menopause status, receptor subtype, tumor laterality, tumor volume, baseline histology grade and treatment regimen (Table 1). Figure 1 shows kinetic parametric maps of three representative cases of a control patient (left column), a pCR patient (center column), and a non-pCR patient (right column). From visual inspection, non-pCRs is more different than pCR and control patients between ipsilateral and contralateral breasts.

Figure 2 shows boxplots of the C-I and I/C comparing between control (blue), pCRs (red), and non-pCRs (orange) groups for parameter (a) *TTMS*, (b) *BAT*, (c) *MaxSlope*,

(d) $AUC30$, (e) K^{trans} , and (f) v_e . Comparing with control and pCRs groups, the non-pCR group had higher I/C ratios and larger C-I for all parameters, except for v_e . Kruskal-Wallis test showed there was significant difference ($p < 0.05$) between pCR and non-pCR groups for K^{trans} and v_e and a statistically significant trend for $TTMS$ ($p = 0.05$) and $AUC30$ ($p = 0.09$), but no significant difference ($p > 0.05$) between control and pCR groups.

Figure 3 shows the ROC curves for three parameters selected by the Lasso regression and for a model that consists of K^{trans} /I/C ratio, v_e I/C ratio and $AUC30$ /I/C ratio with Youden index marked (circle) on the curve. The parameter K^{trans} and $AUC30$ has the largest and the smallest area under the curve, respectively. Finally, Table 2 summarizes the optimum threshold, diagnostic performance and univariable ORs of the six kBPE parameters used to predict pCR. Only K^{trans} /I/C ratio of more than 1.11 (OR = 11 [95% CI: 1.7, 70.0], $p = .015$) and v_e I/C ratio of less than 1.10 (OR = 14, [95% CI: 2.1, 94.2], $p = .035$) were significantly associated with pCR. The sensitivity and specificity of the K^{trans} /I/C ratio cutoff were 83.3% [95% CI: 64.8, 100], and 75% [95% CI: 57.4, 92.6], respectively. The sensitivity and specificity of the v_e I/C ratio cutoff were 75% [95% CI: 57.3, 92.7] and 81.5% [95% CI: 63.0, 99.5].

4. Discussion

While HER2+ breast cancer are considered good responders to neoadjuvant treatment, the rate of patients achieving pCR is still low [3]. Accurate pre-treatment prediction of response to NAC may help determine optimal treatment, lower the overall cost and spare exposure to toxicity from unnecessary chemotherapy treatment [18]. Our study showed that the bilateral asymmetry kBPE parameters K^{trans} /I/C ratio and V_e I/C ratio of top 10% enhancing parenchymal voxels measured from pre-NAC ultrafast DCE-MRI discriminated between pCR and non-pCR in patients with HER2+ breast cancer ($p < 0.05$). Breast cancer patients who achieved pCR after NAC showed lower I/C ratio in $MaxSlope$, $AUC30$, and K^{trans} , and smaller C-I difference in $TTMS$ and BAT between two breasts in pre-NAC. A large difference in BPE kinetics between the affected and contralateral breast – as measured in pre-NAC MRI scans – is strongly associated with residual disease following completion of NAC.

Breast cancer invasiveness is strongly correlated with angiogenesis [35]. Increased vascularity in one breast, presumably due to secretion of angiogenic factors by the cancer, has been used as a diagnostic marker for ipsilateral invasive breast cancer in many studies [36, 37]. However, quantitative assessment of tumor-associated angiogenesis or vascular asymmetry is a challenging task that requires high accuracy of MRI. Increased signal enhancement in ipsilateral normal parenchyma that is associated with elevated blood supply and early uptake of contrast media in ipsilateral breast can be detected more easily by ultrafast DCE-MRI combined with the quantitative analysis described here and may be used as a marker of angiogenesis.

Many of the previous work investigating BPE (as opposed to the kinetic parameters used here) as a predictor of treatment response in breast cancer measured BPE unilaterally in one breast, either in the ipsilateral [22] or contralateral breast [10, 21, 38, 39], or

averaged BPE from both breasts [20, 24, 40]. However, our earlier study [7] and others [19] demonstrated that the pre-NAC BPE or kBPEs from one breast or averaged over two breasts can vary significantly between individuals and is strongly associated with hormonal *milieu* and cardiac function. The ratio or difference between the ipsi-and contra-lateral kBPE parameters can provide much more effective diagnostic and prognostic markers as it corrects for physiologic variations.

In this study, we performed both semi-quantitative and quantitative pharmacokinetic analysis for pre-NAC ultrafast DCE-MRI, obtaining four semi-quantitative kBPE parameters (*MaxSlope*, *AUC30*, *TTMS* and *BAT*) and two quantitative kBPE parameters (K^{trans} and v_e). KBPE asymmetry parameters – I/C ratios of K^{trans} and v_e – are significantly different between pCRs and non-pCRs ($p < 0.05$). However, we observed no evidence of strong correlation between *MaxSlope* I/C ratio, *AUC30* I/C ratio, *TTMS* C-I, *BAT* C-I ($p \geq 0.05$) and pCR. In addition, since v_e is important, following washout for a longer period might be important to measure v_e more accurately.

An accurate BAT measurement relies heavily on the temporal resolution of DCE-MRI. Our earlier study [7] demonstrated that pCR patients have significantly smaller difference of pre-NAC BAT between two breasts than the patients with residual disease. In this study, a similar trend is observed but the BAT difference between two breasts is not significantly different between pCRs and non-pCRs. This could be due to the fact that this study includes more patients who underwent ultrafast DCE-MRI with relatively lower temporal resolution (ultrafast temporal resolution = 3 s (16), 7 s (9), 9 s (3)) than the earlier study (ultrafast temporal resolution = 3 s (17), 7 s (6), 9 s (1)). The average difference of BAT between the two breasts is as small as 1 s. As a result, even with a temporal resolution of 3 seconds and a fitting model, it is challenging to accurately measure BAT changes in individuals. Increasing the sampling rate of DCE-MRI has potential of improving the diagnostic and prognostic accuracy [41].

Limitations of this study include its small sample size and retrospective nature. First, the study included only a modest number of patients with HER2+ breast cancer. Second, the temporal resolution of ultrafast DCE-MRI ranged from 3 – 9 seconds per image and more than 40% of the inclusive patients underwent ultrafast DCE-MRI with relatively low temporal resolution (≥ 7 seconds/phase), which lowers the accuracy of BAT measurements. Third, T1 mapping was not available in this retrospective study, and this could limit the accuracy of pharmacokinetic modeling. Finally, a fixed threshold (top 10%) was used in this study, which could show different levels of voxels enhancement between patients. In future work, artificial intelligent should be consider deriving optimal cutoff for different patients with large sample size.

5. Conclusion

In conclusion, bilateral kBPE parameters K^{trans} I/C ratio and V_e I/C ratio measurement from the initial uptake phase of ultrafast DCE-MRI can be independent predictive factors of pCR in patients with HER2+ breast cancer. Ongoing studies are underway to validate our results in a larger cohort with more breast cancer subtypes.

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Highlights

- K^{trans} I/C ratio and V_e I/C ratio measurement from the initial uptake phase of ultrafast DCE-MRI can be independent predictive factors of pCR in patients with HER2+ breast cancer.
- A large difference in BPE kinetics between the affected and contralateral breast as measured in pre-NAC MRI scans is strongly associated with residual disease following completion of NAC.
- The ratio or difference between the ipsi- and contra-lateral kBPE parameters can provide much more effective diagnostic and prognostic markers as it corrects for physiologic variations.

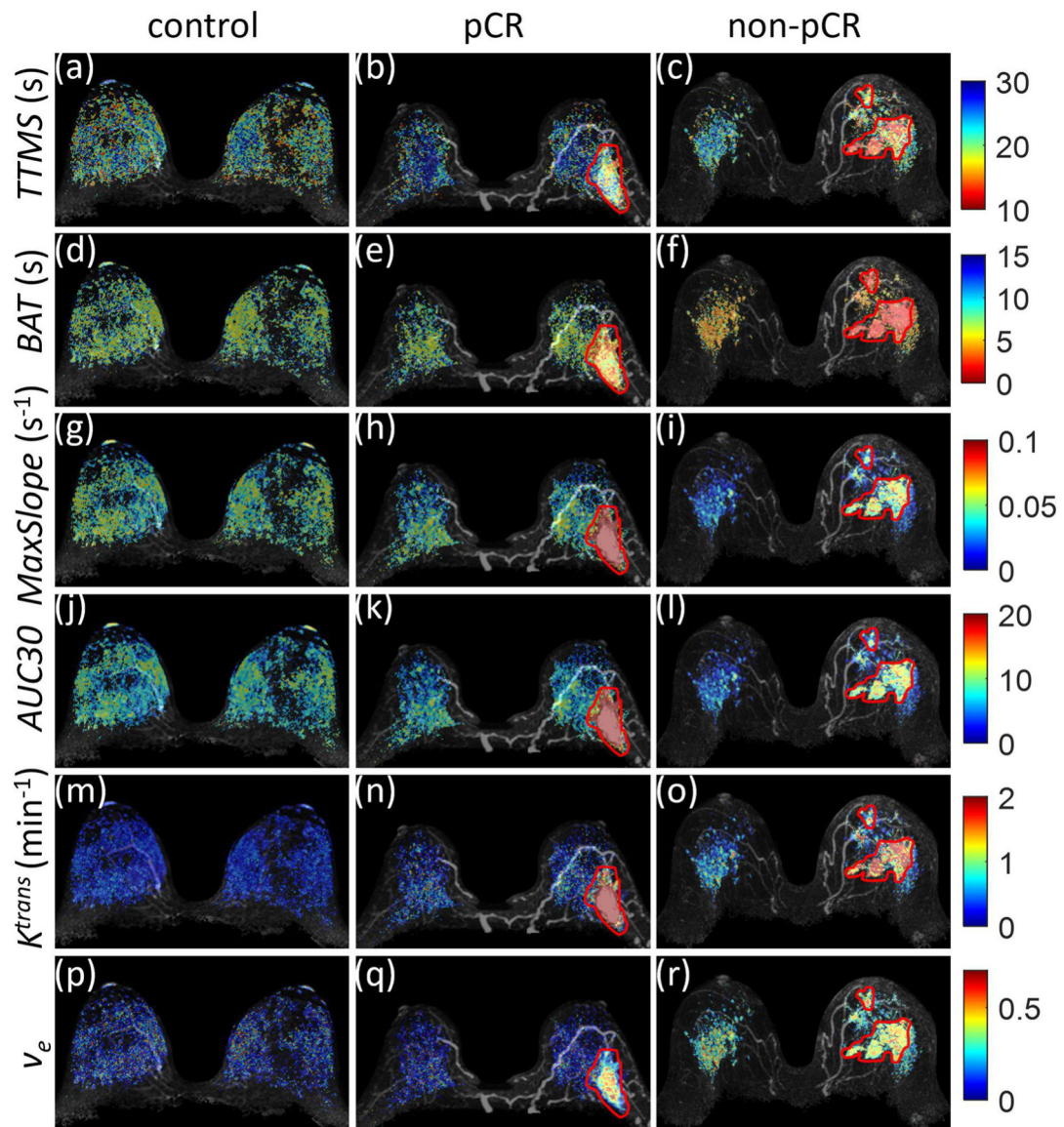


Figure 1.

Axial maximum value projections (MVP) or minimum value projection (mVP) of kinetic parametric maps are shown for a control case (left column: a 40-year-old screening patient), a pCR case (center column: a 47-year-old patient diagnosed with an IDC grade 3 in the left breast who achieved pCR) and a non-pCR case (right column: a 49-year-old patient diagnosed with an IDC grade 3 in the left breast who achieved RCB-I). (a) – (c) axial mVP of *TTMS* map, (d) – (f) axial mVP of *BAT* map, (g) – (i) axial MVP of *MaxSlope* map, (j) – (l) MVP of *AUC30* map, (m) – (o) MVP of K^{trans} map, (p) – (r) MVP of v_e map. The tumor area is circled with red. The window of kinetic parametric mapping is set according to kBPE, as a result, the color is saturated in tumor for some parameters, such as *MaxSlope*, *AUC30*, K^{trans} in the pCR patient.

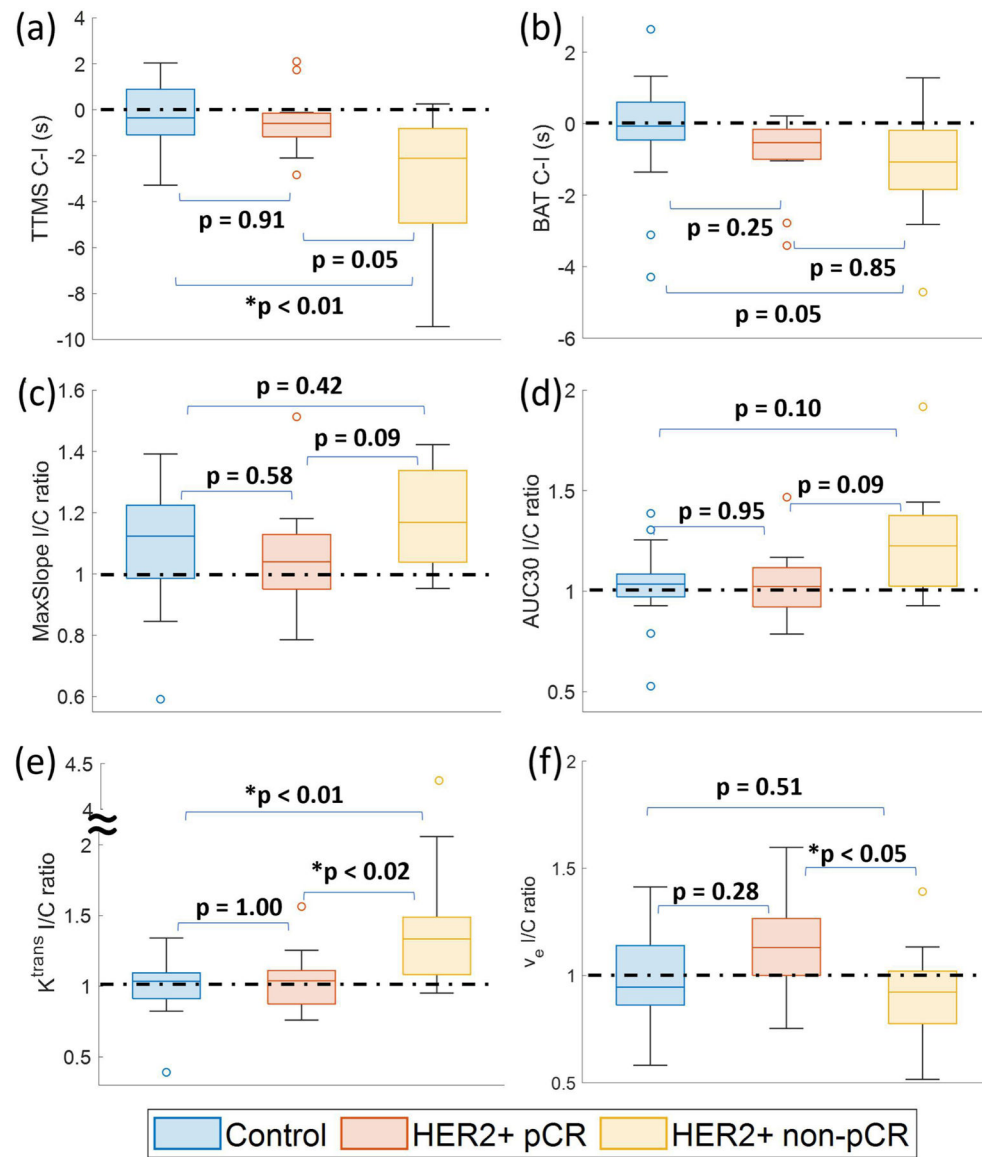


Figure 2. Box-plots of calculated all kBPE asymmetry parameters of control cases (blue) vs. patients with (orange) and without (yellow) pCR for: (a) *TTMS*, (b) *BAT*, (c) *MaxSlope*, (d) *AUC30*, (e) K^{trans} , (f) v_e . The * indicates calculated p-value was statistically significant.

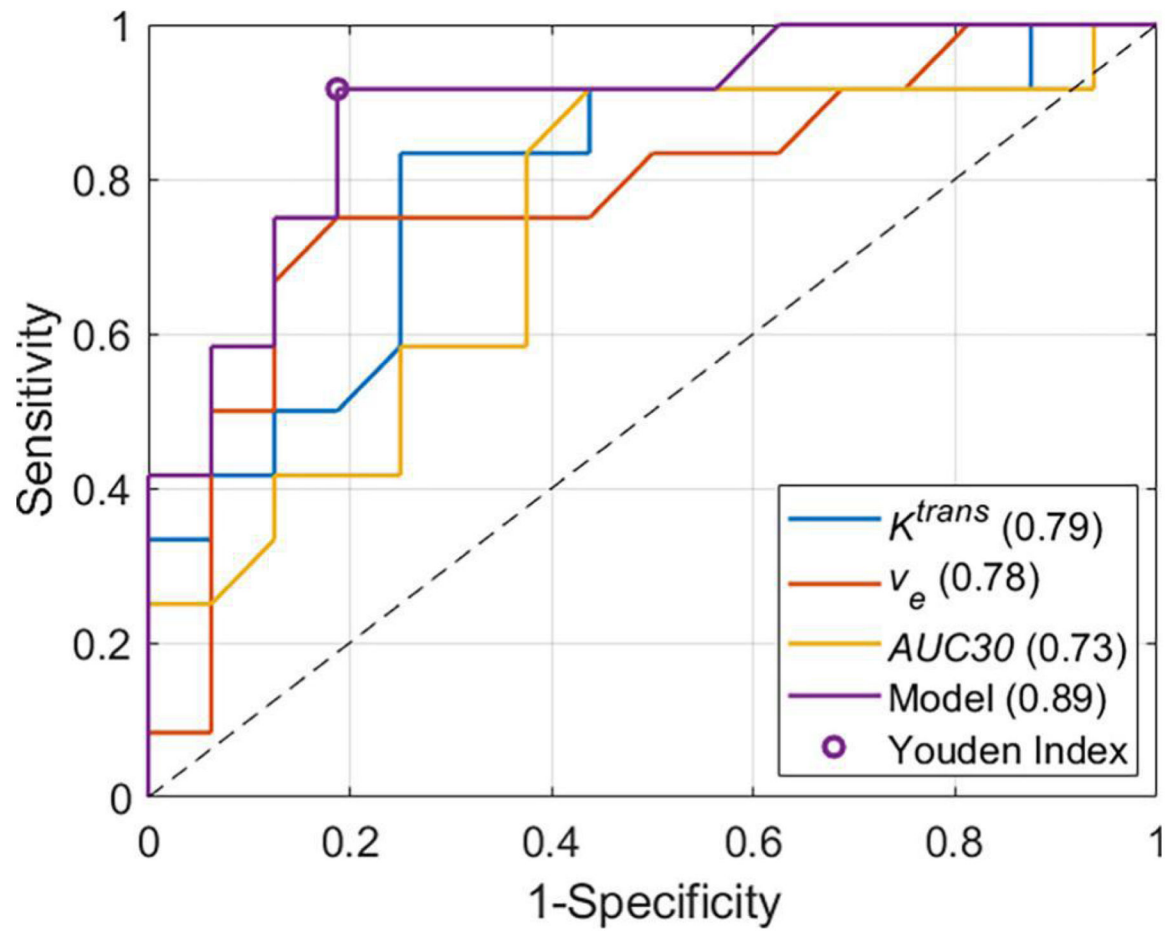


Figure 3.

Graph shows receiver operating characteristic (ROC) curves for the three-kBPE model for $AUC30$ I/C ratio (yellow line), K^{trans} I/C ratio (blue line) and v_e I/C ratio (orange line). The ROC curve for model was shown in purple line with Youden index marked (circle) on the curve.

Table 1.

Baseline Characteristics of the study sample

	Overall (n = 28)	Residual Disease (n = 16)	pCR (n = 12)	p-value
Age, median (range)	53 (25 – 71)	58 (25 – 71)	43 (30 – 65)	.053
Race, n (%)				.78
White	14 (50)	8 (50)	6 (50)	
African American	9 (32)	6 (38)	3 (25)	
Asian / Indian	2 (7)	1 (6)	1 (8)	
More than one race	3 (11)	1 (6)	2 (17)	
Menopause Status, n (%)				.06
Premenopausal	14 (50)	5 (31)	9 (75)	
Postmenopausal	14 (50)	11 (69)	3 (25)	
Receptor Subtype, n (%)				.77
HR+/HER2+	19 (68)	11 (69)	8 (67)	
HR-/HER2+	9 (32)	5 (31)	4 (33)	
Laterality, n (%)				.48
Left	13 (46)	6 (38)	7 (58)	
Right	15 (54)	10 (62)	5 (42)	
Histology				
Invasive Ductal Carcinoma				
Grade, n (%)				.82
2	4 (14)	3 (19)	1 (8)	
3	24 (86)	13 (81)	11 (92)	
Tumor volume (cm³)				
Median (range)	20.6 (5.5 – 552.4)	15.0 (5.5 – 552.4)	27.2 (7.9 – 234.4)	.88
Treatment, n (%)				.39
Taxane + Anthracycline	6 (21)	2 (13)	4 (33)	
Trastuzumab containing	22 (79)	14 (787)	8 (67)	
Pathologic response, n (%)				N/A
pCR	12 (43)		12 (100)	
RCB-I	7 (25)	7 (44)		
RCB-II	9 (32)	9 (56)		

Note:

Data are numbers of patients, with percentages in parentheses. P values were obtained using the chi-squared test and two-sided t-test. HER2+ = human epidermal growth factor receptor 2 positive, pCR

Table 2

Diagnostic performance and univariable analysis of optimal thresholds of ultrafast DCE-MRI kBPEs for predicting pCR

kBPE Parameters	Optimal Threshold	Sensitivity	Specificity	Odds Ratio
<i>TTMSC</i> -I (s)	-1.33	83.3 [64.8, 100]	75 [57.4, 92.6]	15 [2.3, 99.6]
<i>BAT</i> C-I (s)	-1.04	83.3 [65.0, 100]	50 [35.5, 64.5]	5 [0.8, 30.5]
<i>MaxSlope</i> I/C ratio	1.18	91.7 [72.4, 100]	50 [35.4, 64.6]	11 [1.1, 106]
<i>AUC30</i> I/C ratio	1.17	91.7 [72.4, 100]	56.3 [40.8, 71.7]	14.1 [1.5, 83.1]
<i>K^{trans}</i> I/C ratio	1.11	83.3 [64.8, 100]	75 [57.4, 92.6]	11 [1.7, 70.0]
<i>v_e</i> I/C ratio	1.10	75 [57.3, 92.7]	81.3 [63.0, 99.5]	14 [2.1, 94.2]

Note:

Data are percentage with 95% CIs in brackets. *MaxSlope* = maximum enhancement slope, *AUC30* = area under the signal enhancement curve for 30 seconds post-contrast, *BAT* = bolus arrival time, *TTMS* = time to maximum slope, pCR = pathologic complete response. I/C ratio = kBPE mean in ipsilateral normal parenchyma/kBPE mean in contralateral parenchyma, C-I = kBPE mean in ipsilateral normal parenchyma – kBPE mean in contralateral parenchyma